Signals and Systems: midterm practice

Time allowed: 45 minutes · Total: 37 points · Version 3786

Paper code: **101**

Materials after the last question

PART I. SIGNALS (12 POINTS)

Question 1. Which signal is periodic? (1 point)

A. x(t) = e^{-t}

B. x(t) = $\cos(4\pi t)$

C. x(t) = $t\,u(t)$

D. x(t) = $\delta(t-1)$

ANSWER

B

Question 2. True or false: an even signal satisfies x(t) = x(−t). (1 point)

T. True F. False

ANSWER

T

Question 3. What is the fundamental period, in seconds, of x(t) = sin(5πt)? (1 point)

Answer: ____

ANSWER

0.4 (± 0.001)

Question 4. Fill in the blank. A system whose output depends only on present and past inputs is called _____. (1 point)

ANSWER

(1) causal

Question 5. Match each signal with its Fourier transform. (2 points)

1. rect(t)	a. 2π·δ(ω)
2. 1	b. sinc(ω / 2π)
3. δ(t)	c. 1

1 → _____ 2 → _____ 3 → _____

ANSWER

1 → b, 2 → a, 3 → c (partial: each pair earns an equal share of the 2 points)

Question 6. Which statement about a Fourier series is correct? (1 point)

A. It holds for a Fourier series in general

B. It never holds for a Fourier series

C. It holds only when the signal is periodic

D. It depends on the sampling rate alone

ANSWER

C

Question 7. Which statement about aliasing is correct? (1 point)

A. It holds for aliasing in general
B. It never holds for aliasing
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

D

Question 8. True or false: statement 8 about a linear system holds. (1 point)

T. True F. False

ANSWER

T

Question 9. Which statement about sampling is correct? (1 point)

A. It holds for sampling in general
B. It never holds for sampling
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

B

Question 10. Name the effect when frequencies above fs/2 fold back into the band. (1 point)

ANSWER

aliasing

Question 11. Which statement about an LTI system is correct? (1 point)

A. It holds for an LTI system in general
B. It never holds for an LTI system
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

D

PART II. SYSTEMS (13 POINTS)

Question 12. Which statement about the Laplace transform is correct? (1 point)

A. It holds for the Laplace transform in general
B. It never holds for the Laplace transform
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

A

Question 13. Which statement about a stable filter is correct? (1 point)

A. It holds for a stable filter in general
B. It never holds for a stable filter
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

B

Question 14. True or false: statement 14 about a linear system holds. (1 point)

T. True F. False

ANSWER

T

Question 15. Put the steps of designing a digital filter in order. (2 points)

A. State the specification
B. Check the response
C. Choose the filter type
D. Compute the coefficients

1 ☐ **2** ☐ **3** ☐ **4** ☐

ANSWER

A, C, D, B (partial: each item in its place earns an equal share of the 2 points)

Question 16. Which statement about a Fourier series is correct? (1 point)

A. It holds for a Fourier series in general
B. It never holds for a Fourier series
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

A

Question 17. Which statement about aliasing is correct? (1 point)

A. It holds for aliasing in general
B. It never holds for aliasing
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

B

Question 18. Which of these are conditions of the sampling theorem? (2 points)

Choose all that apply.

A. The signal is band-limited
B. fs is more than twice the highest frequency
C. The signal is periodic
D. The samples are quantised to 16 bits

ANSWER

A, B (partial: each correct choice earns an equal share of the 2 points; each wrong choice takes one share back)

Question 19. Which statement about sampling is correct? (1 point)

A. It holds for sampling in general
B. It never holds for sampling
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

D

Question 20. Readings (2 points)

Gain (dB) _____

Cutoff (Hz) _____

ANSWER

Gain (dB): 3

Cutoff (Hz): 100 (partial: each field earns an equal share of the 2 points)

Question 21. What does gain(2.5) return, to three decimals? (1 point)

Answer: [ ]

ANSWER

0.371 (± 0.001)

PART III. SAMPLING (12 POINTS)

Question 22. Which statement about the Laplace transform is correct? (1 point)

A. It holds for the Laplace transform in general
B. It never holds for the Laplace transform
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

C

Question 23. True or false: statement 23 about a linear system holds. (1 point)

T. True F. False

ANSWER

T

Question 24. How sure are you of your answers so far? (1 point)

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Not at all — Very sure

ANSWER

Not marked

Question 25. Which statement about convolution is correct? (1 point)

A. It holds for convolution in general
B. It never holds for convolution
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

B

Question 26. Which statement about a Fourier series is correct? (1 point)

A. It holds for a Fourier series in general
B. It never holds for a Fourier series
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

C

Question 27. Explain why sampling below the Nyquist rate causes aliasing. (4 points)

ANSWER

See the rubric

RUBRIC

Criterion	Points
States fs ≥ 2·fmax	1
Explains that spectral copies overlap	2
Gives a worked example	1

Question 28. Which statement about a causal system is correct? (1 point)

A. It holds for a causal system in general
B. It never holds for a causal system
C. It holds only when the signal is periodic
D. It depends on the sampling rate alone

ANSWER

A

Question 29. Which statement about sampling is correct? (1 point)

A. It holds for sampling in general

B. It never holds for sampling

C. It holds only when the signal is periodic

D. It depends on the sampling rate alone

> **ANSWER**
>
> B

Question 30. True or false: statement 30 about a linear system holds. (1 point)

T. True F. False

> **ANSWER**
>
> T

You may detach this page. It holds no questions.

FOURIER TRANSFORM PROPERTIES

Property	x(t)	X(ω)	Condition or note
Linearity	$a\,x(t) + b\,y(t)$	$a\,X(\omega) + b\,Y(\omega)$	a and b constants, both signals have transforms
Time shift	$x(t - t_0)$	$e^{-j\omega t_0}X(\omega)$	any real shift t0
Frequency shift	$e^{j\omega_0 t}x(t)$	$X(\omega - \omega_0)$	any real shift omega0
Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X(\omega/a)$	a is real and not zero
Convolution	$x(t) * y(t)$	$X(\omega)\,Y(\omega)$	both signals absolutely integrable
Multiplication	$x(t)\,y(t)$	$\frac{1}{2\pi}X(\omega) * Y(\omega)$	dual of convolution in time

Parseval: $\int \lvert x(t)\rvert^2 dt = \frac{1}{2\pi}\int \lvert X(\omega)\rvert^2 d\omega$.

$\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$, so sinc(0) = 1.

TRIGONOMETRIC IDENTITIES

- $e^{j\theta} = \cos\theta + j\sin\theta$
- $\cos\theta = (e^{j\theta} + e^{-j\theta})/2$
- $\sin\theta = (e^{j\theta} - e^{-j\theta})/2j$
- $\cos a\cos b = \frac{1}{2}[\cos(a-b) + \cos(a+b)]$
- $\sin^2\theta = (1 - \cos 2\theta)/2$

UNITS AND CONSTANTS

Quantity	Value
Angular and ordinary frequency	$\omega = 2\pi f$
Gain in decibels	$20\log_{10}\lvert H\rvert$
-3 dB as a ratio	$1/\sqrt{2} \approx 0.7071$
Pi	3.14159